Research Summary

Alloxazine Derivatives as Tunable PCET Mediators for Electrochemical pH-Swing CO_2 Capture

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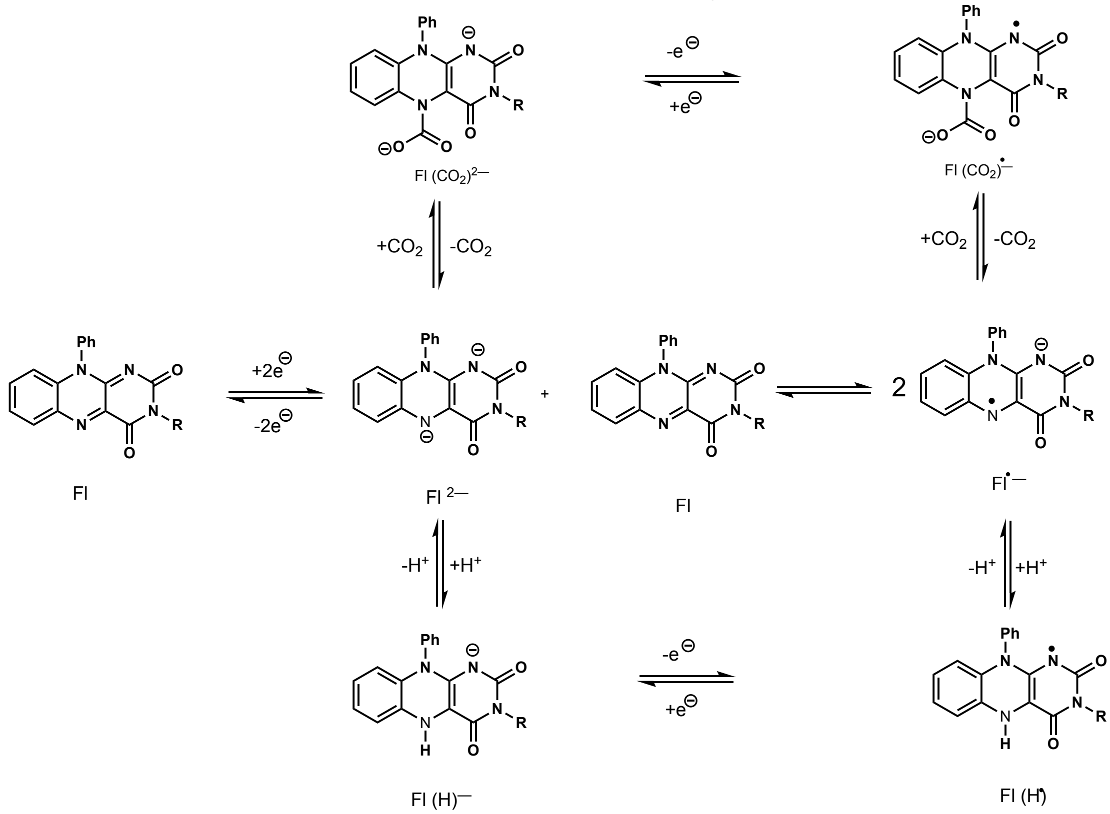
Background & Motivation

Electrochemical pH-swing strategies offer a low-energy route to CO_2 capture: alkaline conditions favor CO_2 absorption as bicarbonate or carbonate, while acidic conditions drive its release, with the requisite pH modulation achieved electrochemically. Central to this approach are proton-coupled electron transfer (PCET) mediators: redox-active molecules that exchange both electrons and protons during cycling to generate the necessary pH swings. Quinone- and phenazine-based mediators have established proof of concept, yet practical limitations persist, including oxygen sensitivity, poor aqueous solubility, and degradation under extended electrochemical operation. Flavins (alloxazines), whose $2e^-/2H^+$ PCET reactivity has been refined by nature across countless enzymatic systems, constitute an underexplored but compelling scaffold for this application.

Scheme 1. Potential electrochemical and chemical pathways available to Flavin derivatives.



Approach & Design

We synthesized three water-soluble alloxazine derivatives (Fl1–Fl3) bearing charged N-substituents designed to enhance aqueous solubility. Fl1 bears a propyl sulfonate, Fl2 carries a trimethylammonium cation, and Fl3 features a hydroxypropyl trimethylammonium cation (combining an OH group with a charged tail). Fl1 serves as a benchmarking control, while F3 is designed to examine whether a proximal hydroxyl can modulate redox behavior through intramolecular H-bonding. All derivatives were characterized by $^{1}H/^{31}C$ NMR and HRMS, and studied via cyclic voltammetry under N_2, and CO_2 atmospheres.

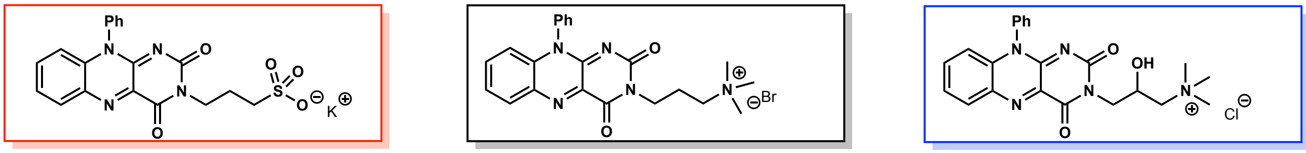


Chart 1. Molecular structures of alloxazine derivatives Fl1–Fl3.

Key Findings

Reversible PCET Redox Chemistry

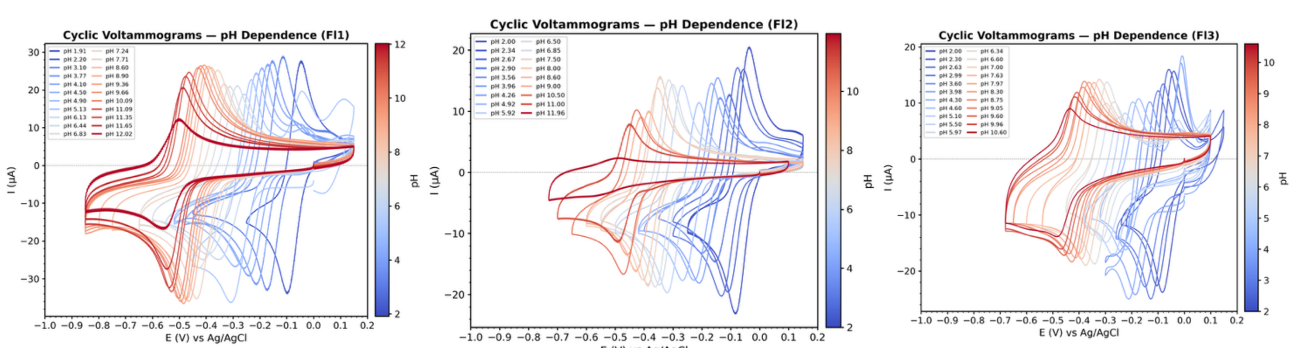
All three derivatives display a single reversible 2e⁻ redox couple in aqueous 1 M KCl: $E_{1/2}$ = –0.41 V (Fl1), –0.41 V (Fl2), and –0.36 V (Fl3) vs Ag/AgCl (Table1). The identical potentials of Fl1 and Fl3 confirm that swapping a distal cation for an anion has negligible effect on the redox-active core. Fl2's ~70 mV anodic shift is attributed to its hydroxyl group engaging in H-bonding with the isoalloxazine ring, stabilizing the reduced form. Furthermore, presence of the group also helps with improving solubility.

Derivative	Solubility (1M KCl)	$E_{1/2}$(Ag/AgCl/ 1M KCl)	pH swing (saturated CO_2)
Fl 1	<< 0.17 M	-0.45 V	5.20 → 10.40
Fl 2	0.17M	-0.46 V	6.50 → 8.56
Fl 3	0.20 M	-0.36 V	6.50 →9.10

Table 1. Solubility, redox potentials and pH swing summary

Pourbaix Analysis & PCET Mechanism

pH-dependent CV and Pourbaix diagrams reveal slopes of –56 to –60 mV/pH below pH ~7, consistent with a $2e^-/2H^+$ transfer mechanism to form FlH_2. Above pH ~7, the slope decreases to ~28–34 mV/pH, indicating a transition to $2e^-/1H^+$ transfer to for $(FlH)^-$. The pK_a transition points are: Fl1 (7.2), Fl2 (7.15), and Fl3 (6.37). While the pK_a of the protonated two electron reduced species are easily measured using electrochemical method, pK_a of singly reduced semiquinones are difficult to measure due to fast disproportionation reaction. As a result, we sought to investigate the nature of the singly reduced product and measure its' pK_a *via* pulse radiolysis experiment.



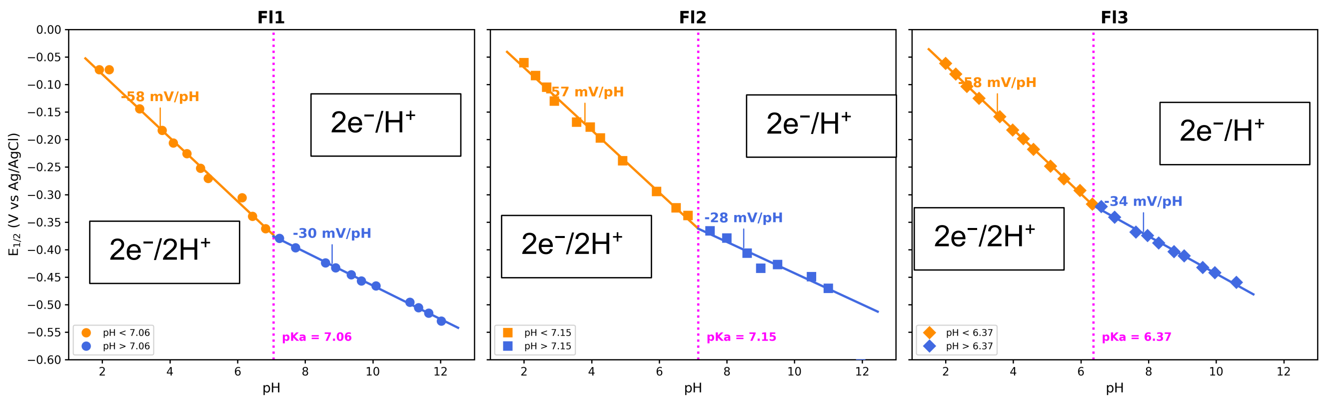


Figure 1. CVs of Fl1–Fl3 in BR buffer (top); Pourbaix diagram for Fl1 (bottom).

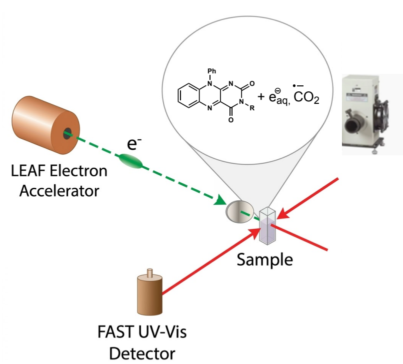
Compound	$E_1/_2$ (V)	Slope (< pH 7)	Slope (> pH 7)	pK_a
Fl1	–0.41	–56.8 mV/pH	–28.4 mV/pH	7.2
Fl2	–0.34	–56.3 mV/pH	–28 mV/pH	7.15
Fl3	–0.41	–58 mV/pH	–34 mV/pH	6.37

Table 1. Electrochemical Parameters and pka of FlH$_2$ for Fl1–Fl3

Potentials vs Ag/AgCl (1 M KCl) in aqueous solution.

Ongoing Work: Pulse Radiolysis Studies

To gain deeper mechanistic insight into the radical intermediates formed during alloxazine reduction, we are conducting pulse radiolysis experiments. Pulse radiolysis enables direct generation and time-resolved spectroscopic observation of transient radical species (semiquinone intermediates) that are difficult to isolate by conventional methods. These measurements provide difference absorption spectra of the one-electron reduced (semiquinone) forms of Fl1–Fl3, along with kinetic data for their formation and decay. The spectra below represent preliminary results from these ongoing experiments.



$$\mathrm{H_2O} \rightsquigarrow e^-_{aq}\ (G = 2.60) + \mathrm{H}^\bullet\ (G = 0.55) + {}^\bullet\mathrm{OH}\ (G = 2.65) + \mathrm{H_2}\ (G = 0.45) + \mathrm{H_2O_2}\ (G = 0.70) \quad (1)$$

$${}^\bullet\mathrm{OH} + \mathrm{HCOO^-} \longrightarrow \mathrm{COO^{\bullet -}} + \mathrm{H_2O} \quad (2)$$

$$\mathrm{H}^\bullet + \mathrm{HCOO^-} \longrightarrow \mathrm{H_2} + \mathrm{COO^{\bullet -}} \quad (3)$$

$$e^-_{aq} + \mathrm{Fl} \longrightarrow \mathrm{Fl^{\bullet -}} \quad (4)$$

$$\mathrm{COO^{\bullet -}} + \mathrm{Fl} \longrightarrow \mathrm{Fl^{\bullet -}} \quad (5)$$

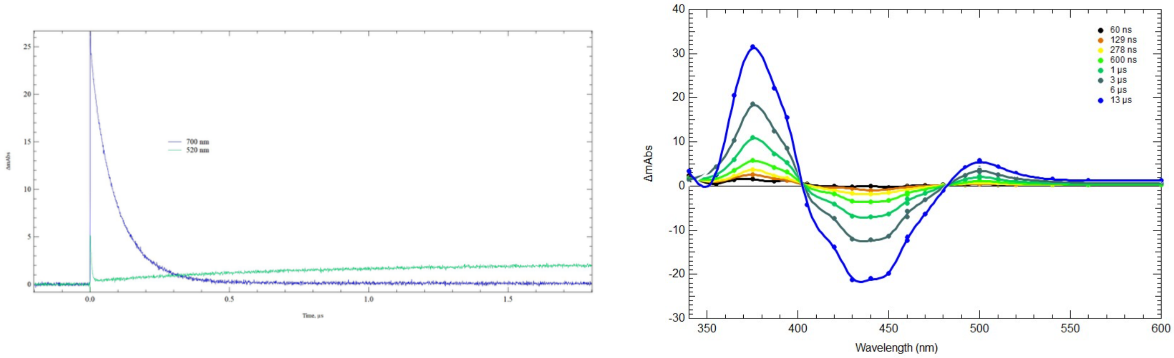


Figure 2. Pulse radiolysis transient absorption spectra of Fl1 semiquinone intermediates and Kinetic traces showing semiquinone formation/decay at selected wavelengths.

Significance & Next Steps

This work establishes alloxazine derivatives as a versatile and tunable molecular platform for electrochemical pH-swing CO_2 capture. Key advantages include: well-defined $2e^-/2H^+$ PCET with predictable pH swings, straightforward N-substitution chemistry for tuning solubility and redox potential, Upcoming goals include: completing pulse radiolysis characterization of radical intermediates, integration into flow cell architectures for continuous capture–release cycling, long-term stability assessment, and computational studies to deconvolute pH-mediated pathways from direct CO_2 binding.